Daiki Shiono

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Education

Tohoku University Apr 2024 – Mar 2027 **(expected)**
Ph.D. Program, Graduate School of Information Sciences
System Information Sciences; Suzuki Laboratory (TohokuNLP Group; Advisor: Prof. Jun Suzuki)

Tohoku University Apr 2022 – Mar 2024
M.S. in Information Science, Graduate School of Information Sciences
System Information Sciences; Suzuki Laboratory (TohokuNLP Group; Advisor: Prof. Jun Suzuki)

Tohoku University Apr 2018 – Mar 2022
B.E. in Engineering, School of Engineering
Department of Electrical, Information and Physics Engineering

Research and Industry Experience

Departmental Visiting Student Sep 2026 – Present

University of Cambridge | Department of Computer Science and Technology
Conducting research in Vision & Language and NLP, hosted by Prof. Andreas Vlachos and collaborating closely with Diana Galván Sosa.

Machine Learning R&D Engineer (Intern / Contractor) Apr 2024 – Oct 2026

Turing Inc. | Fundamental AI Group
Contributed to training large-scale vision-language models (LVLMs), building a large-scale Japanese vision-language pretraining dataset, and constructing a Japanese visual instruction-tuning dataset.

Research Assistant Apr 2023 – Mar 2027 (expected)

Center for Data-driven Science and Artificial Intelligence, Tohoku University
Supported research activities through experimental setup, documentation, data maintenance, and related project operations.

Part-time Machine Learning Engineer Mar 2023 – Dec 2023

Future Inc. | Strategic AI Group
Improved OCR models and developed a model for converting foreign personal names into Japanese readings.

Research Support Programs and Scholarships

Research Fellowship Sep 2024 – Mar 2027 (expected)

BOOST Program (Next-Generation AI Talent Development)
Received a research stipend and research funding through the Japan Science and Technology Agency (JST) BOOST initiative.

Research Fellowship Apr 2020 – Mar 2025

WISE Program for AI Electronics (AIE), Tohoku University
Participated in this MEXT-supported graduate education program and received educational and research support funding.

Scholarship Apr 2022 – Mar 2024

Japan Student Services Organization (JASSO) Scholarship
Granted a 50% repayment exemption in recognition of outstanding academic achievements.

Teaching Experience

Teaching Assistant Oct 2023 – Feb 2024

Natural Language Processing
Graduate-level course, Graduate School of Information Sciences, Tohoku University

Teaching Assistant Oct 2022 – Dec 2023

Introduction to Programming for Artificial Intelligence Technology
General education course, Tohoku University

Projects

Building a Large-Scale Collection of SFT Datasets Translated into Japanese Link to Page

Independent project
Ongoing

Large-Scale Fine-Tuning of a Japanese LLM for Mathematical Reasoning Link to Page

Collaborated on the large-scale fine-tuning of a Japanese LLM specialized for mathematical reasoning tasks, using NeMo on the ABCI 3.0 supercomputer.
Achieved **1st place** in the open division and **2nd place** overall in the FT-LLM2026 Math Task.
March 2026

Sharing Slides Summarizing Insights on LVLMs as of 2024 Link to Slide

Independent project
November 2024

Building a Japanese BERT Model for Long-Sequence Processing Link to Page

Collaborated on the development and distributed pretraining of a Japanese BERT model supporting long-sequence inputs, using Megatron-LM on the TSUBAME4.0 supercomputer.
June 2024

Building SFT Datasets for Japanese LVLMs Link to Page

Outcome from my internship at Turing Inc.
September 2024

Journal Publications

Daiki Shiono, Shingo Yokoi, Shintaro Inuzuka, Tsubasa Takahashi, Jun Suzuki, Yu Yamaguchi. “MOMIJI: 日本語大規模インターリーブ視覚言語データセット.” *Journal of Natural Language Processing*, Volume 33, Issue 4 (General Paper; scheduled for publication December 15, 2026). **Accepted September 18, 2026**.
Acceptance Listing Dataset Code

Daiki Shiono, Shumpei Miyawaki, Ryota Tanaka, Jun Suzuki. “Instruction-Following Evaluation of Large Vision-Language Models.” *New Generation Computing*, Volume 44, Article 7, January 2026.
Link to Paper arXiv Training Data

Shoji Moriya*, **Daiki Shiono***, Riki Fujihara, Yosuke Kishinami, Subaru Kimura, Shusaku Sone, Reina Akama, Yuta Matsumoto, Jun Suzuki, Kentaro Inui (*equal contributions). “Aoba_v3 bot: a multimodal chatbot system combining rules and various response generation models.” *Advanced Robotics*, Volume 37, Issue 21, pp.1392–1405, August 2023.
Link to Paper

International Conference Publications

Raito Kiya, Satoki Ohashi, Kosuke Sato, Go Kamoda, Ryosuke Takahashi, Yuji Yamamoto, **Daiki Shiono**, Keisuke Sakaguchi, Goro Kobayashi. “It’s Not RoPE That Creates Sinks: The Role of Self-Concentration and Value-Non-Mixing in Attention.” *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing* ***(EMNLP 2026)***, Budapest, Hungary, pp.31160–31184, October 2026.
Link to Paper arXiv Code

Futa Waseda, Shojiro Yamabe, **Daiki Shiono**, Kento Sasaki, Tsubasa Takahashi. “Read or Ignore? A Unified Benchmark for Typographic-Attack Robustness and Text Recognition in Vision-Language Models.” *Proceedings of the European Conference on Computer Vision* ***(ECCV 2026)***, Malmö, Sweden, September 2026.
Link to Paper arXiv Project Dataset Code

Shojiro Yamabe, Futa Waseda, **Daiki Shiono**, Tsubasa Takahashi. “Text-Printed Image: Bridging the Image-Text Modality Gap for Text-centric Training of Large Vision-Language Models.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* ***(CVPR 2026)***, Denver, USA, June 2026. **CVPR Compute Gold Star Award**.
Link to Paper arXiv Poster

Keishi Ishihara*, Kento Sasaki*, Tsubasa Takahashi, **Daiki Shiono**, Yu Yamaguchi (*equal contributions). “STRIDE-QA: Visual Question Answering Dataset for Spatiotemporal Reasoning in Urban Driving Scenes.” *Proceedings of the AAAI Conference on Artificial Intelligence* ***(AAAI 2026)***, Singapore, January 2026.
Link to Paper arXiv Project Training Data Benchmark Code

Ko Yoshida, **Daiki Shiono**, Kai Sato, Toko Miura, Momoka Furuhashi, Jun Suzuki. “Batch-wise Convergent Pre-training: Step-by-Step Learning Inspired by Child Language Development.” *Proceedings of the First* ***BabyLM Workshop***, Association for Computational Linguistics, Suzhou, China, pp.508–524, November 2025.
Link to Paper

Daiki Shiono, Ana Brassard, Yukiko Ishizuki, Jun Suzuki. “Evaluating Model Alignment with Human Perception: A Study on Shitsukan in LLMs and LVLMs.” *Proceedings of the 31st International Conference on Computational Linguistics* ***(COLING 2025)***, Abu Dhabi, UAE, January 2025.
Link to Paper Dataset Code

Daiki Shiono, Shumpei Miyawaki, Ryota Tanaka, Jun Suzuki. “Instruction-Following Evaluation for Large Vision-Language Models.” *Non-archival submission for the 2024 Conference of the North American Chapter of the Association for Computational Linguistics Student Research Workshop* ***(NAACL SRW 2024)***, Mexico City, Mexico, June 2024.

Ryohei Kamei, **Daiki Shiono**, Reina Akama, Jun Suzuki. “Detecting Response Generation Not Requiring Factual Judgment.” *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics Student Research Workshop* ***(NAACL SRW 2024)***, Mexico City, Mexico, June 2024.
Link to Paper arXiv

Yukiko Ishizuki*, **Daiki Shiono***, Ana Brassard, Jun Suzuki (*equal contributions). “Exploring Large Language Models’ Understanding of Shitsukan.” *Non-archival submission for the 18th Conference of the European Chapter of the Association for Computational Linguistics Student Research Workshop* ***(EACL SRW 2024)***, Malta, March 2024.

Domestic Conference Presentations

Daiki Shiono, Shingo Yokoi, Shintaro Inuzuka, Tsubasa Takahashi, Jun Suzuki, Yu Yamaguchi. “MOMIJI: 日本語大規模インターリーブ視覚言語データセット.” *言語処理学会第 32 回年次大会* ***(NLP 2026)***, pp.717–722, March 2026.
Link to Paper

Kazuki Inomata, Haruto Yoshida, **Daiki Shiono**, Reina Akama, Keisuke Sakaguchi. “JCuisines: 日本の地域文化の理解にむけた郷土料理視覚言語ベンチマーク.” *言語処理学会第 32 回年次大会 **(NLP 2026)***, pp.711–716, March 2026.
Link to Paper

Satoki Ohashi, Raito Kiya, Kosuke Sato, Ryosuke Takahashi, Go Kamoda, Yuji Yamamoto, **Daiki Shiono**, Keisuke Sakaguchi, Goro Kobayashi. “注意機構における Attention Sink のバイアス項的解釈.” *言語処理学会第 32 回年次大会 **(NLP 2026)***, pp.2591–2596, March 2026. **Young Researcher Encouragement Award**.
Link to Paper

Raito Kiya, Satoki Ohashi, Kosuke Sato, Go Kamoda, Ryosuke Takahashi, Yuji Yamamoto, **Daiki Shiono**, Keisuke Sakaguchi, Goro Kobayashi. “Attention Sink および Massive Activation の発生機序の分解.” *言語処理学会第 32 回年次大会 **(NLP 2026)***, pp.2579–2584, March 2026. **Young Researcher Encouragement Award**.
Link to Paper

Keito Kudo, Masatoshi Hidaka, **Daiki Shiono**, Go Kamoda, Mengyu Ye, Ryosuke Takahashi, Yoichi Aoki, Kazuki Yano, Masaya Taniguchi, Takumi Ito, Keisuke Sakaguchi, Jun Suzuki. “チーム dentaku FT-LLM 2026 (数学タスク) システム解説.” *The Second Workshop on Fine-Tuning and Evaluation of Large Language Models **(FT-LLM 2026)***, Utsunomiya, Tochigi, March 2026. **1st Place (Open Division), 2nd Place (Overall)**.
Link to Paper Poster Slides Models/Datasets Code

Ko Yoshida, **Daiki Shiono**, Kai Sato, Momoka Furuhashi, Toko Miura, Jun Suzuki. “バッチ単位収束型事前学習：子どもの言語発達に着想を得た一歩ずつの学習.” *NLP 若手の会第 20 回シンポジウム **(YANS 2025)***, August 2025. **Encouragement Award**.

Daiki Shiono, Ryota Tanaka, Shumpei Miyawaki, Keito Kudo, Jun Suzuki. “Padding vs. Packing: 大規模言語モデルのファインチューニングにおける学習効果の検証.” *言語処理学会第 31 回年次大会 **(NLP 2025)***, pp.1698–1703, March 2025.
Link to Paper

Ryosuke Matsuda, **Daiki Shiono**, Ana Brassard, Jun Suzuki. “大規模視覚言語モデルの質感知覚能力の分析.” *言語処理学会第 31 回年次大会 **(NLP 2025)***, pp.2550–2555, March 2025.
Link to Paper

Keito Kudo, Go Kamoda, **Daiki Shiono**, Jun Suzuki. “日本語バイト符号化マスク言語モデルの開発と分析.” *言語処理学会第 31 回年次大会 **(NLP 2025)***, pp.3356–3361, March 2025.
Link to Paper ByBERT-JP Models ByGPT-JP Model

Daiki Shiono, Shumpei Miyawaki, Ryota Tanaka, Jun Suzuki. “大規模視覚言語モデルに関する指示追従能力の検証.” *言語処理学会第 30 回年次大会 **(NLP 2024)***, pp.2057–2062, March 2024. **Committee Special Award**.
Link to Paper

Ryohei Kamei, **Daiki Shiono**, Reina Akama, Jun Suzuki. “事実正誤判定が不要な生成応答の検出に向けたデータセットの収集と分析.” *言語処理学会第 30 回年次大会 **(NLP 2024)***, pp.2125–2130, March 2024.
Link to Paper

Ryohei Kamei, **Daiki Shiono**, Reina Akama, Jun Suzuki. “生成応答に含まれる事実に基づかない情報の自動検出の試み.” *NLP 若手の会第 18 回シンポジウム **(YANS 2023)***, August 2023. **Encouragement Award**, **CyberAgent Award**.

Daiki Shiono, Shumpei Miyawaki, Haruki Nagasawa, Jun Suzuki. “画像キャプションを利用した IconQA タスクへのアプローチ.” *言語処理学会第 29 回年次大会 **(NLP 2023)***, pp.1857–1862, March 2023.
Link to Paper

Shoji Moriya*, **Daiki Shiono***, Yosuke Kishinami, Riki Fujihara, Subaru Kimura, Yuta Matsumoto, Shusaku Sone, Reina Akama, Jun Suzuki, Kentaro Inui (*equal contributions). “aoba_v3 bot: 多様な応答生成モデルとルールベースを統合したマルチモーダル雑談対話システム.” *人工知能学会音声・言語理解と対話処理研究会第 96 回研究会 (JSAI SIG-SLUD 96) 第 13 回対話システムシンポジウム*, December 2022. **Excellence Award, 2nd Place, Open Track, 5th Dialogue System Live Competition**.
Link to Paper

Book

Shuhei Kurita, Seitaro Shinagawa, Rintaro Yanagi, **Daiki Shiono**, Shinnosuke Takamichi, Daisuke Niizumi, Tomohiko Nakamura, Takuma Yagi, Takayuki Shinohara. “**マルチモーダルモデル入門 (Introduction to Multimodal Models)**.” Gijutsu-Hyoron Co., Ltd., September 28, 2026. ISBN 978-4-297-15836-1.
Amazon Publisher

Awards

1st Place (Open Division), 2nd Place (Overall), FT-LLM2026 Math Task (March 2026).

Committee Special Award, 30th Annual Meeting of the Association for Natural Language Processing **(NLP 2024)** (26/567 submissions, ≈4.59%) (March 2024).

Excellence Award (2nd Place), Open Track, 5th Dialogue System Live Competition (December 2022).

Talks

“Large Vision-Language Models: How They See, Pass, and Answer.”
NLIP Seminar Series, Department of Computer Science and Technology, University of Cambridge. October 2026.
Link to Seminar Page

“Words or Vision: Do Vision-Language Models Have Blind Faith in Text? (CVPR 2025).”
The 17th Saisentan NLP Study Group (SNLP 2025). August 2025.
Link to Presentation Materials

Skills

Software & Tools Python, PyTorch, Transformers, Linux, Docker, Singularity.